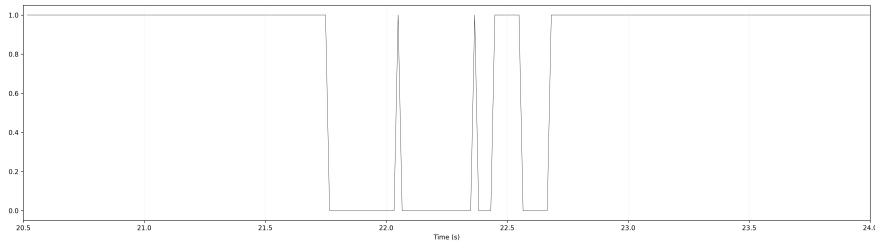
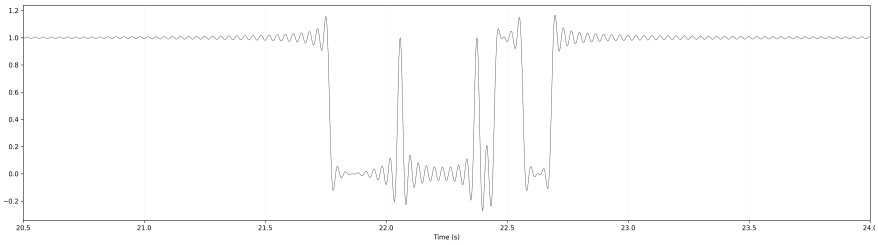
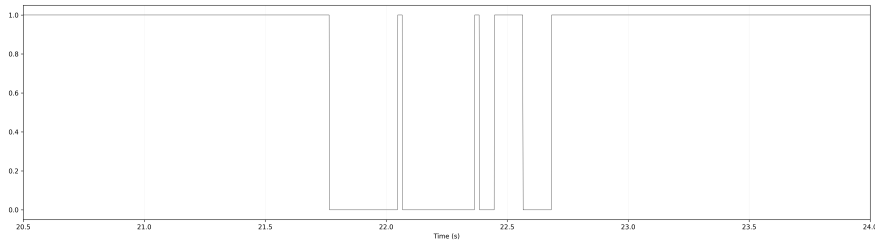
The original mne-bids-eyetracking-pipeline produced non-discrete values for step-like ET channels such as `L Validity`, `R Validity`, and `Pupil Confidence`. This occurred because `mne.preprocessing.realign_raw` resamples all channels as continuous signals during clock-drift correction. The applied fix in the new pipeline was to round back the affected channels to the nearest integer at each sample after realignment.



(a) Original recording



(b) Resampled



(c) Rounded

Figure 1 – A sample segment of the `L Validity` signal. The plotted lines connect consecutive samples with diagonal lines. The step-like signals only contain `0` and `1`

This correction removes the resampling artifacts, but it does not exactly correspond to the original signal. For example, the single-sample `1` at 22.05s in the original recording lasts about 16.7 ms, while the corresponding spike in the rounded 500 Hz signal spans 10 samples, or 20 ms. A more exact approach would correctly handle these channels within `realign_raw`, rather than correcting them afterwards.